

ShopSphere

S11 - Coupon Rules and Pricing Engine

Backend + DB

Platform-Specific Sprint Planning Document

This document intentionally defines only the behavior and delivery scope of the selected platform. It does not mix implementation details from the other platforms.

1. Sprint Definition

Sprint: S11 - Coupon Rules and Pricing Engine

Platform: Backend + DB

Application: ShopSphere

Objective: Implement the backend and database boundary for Coupon Rules and Pricing Engine, including business rules, persistence and API behavior.

This sprint is complete only when the stated use cases, acceptance criteria and tests pass for this platform.

2. Business Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Expected Result
UC-01	Create coupon	Marketing admin	Authorized role	Configure rules and activate	Coupon becomes valid within configured window.
UC-02	Apply coupon	Customer	Cart and user satisfy rules	Submit coupon	Eligible discount is applied once.

3. Detailed Implementation Plan

1. Implement coupon types, targeting, minimum order, maximum discount and user limits.
2. Implement validity windows and usage counters.
3. Build transactional coupon application and authoritative pricing breakdown.

4. Data / State / Contract Scope

- coupons
- coupon_rules
- coupon_usage

API / Integration Contract

- Feature APIs for this sprint, versioned under /api/v1.
- Every endpoint must define authentication, authorization, validation, error codes and request correlation.

5. Rules and Expected Behavior

- The platform must rely on the backend as the source of truth for business rules.
- Do not expose stack traces, secrets or internal implementation details to end users.
- Handle loading, empty, validation and error states explicitly.
- Requests that can be retried must be safe from duplicate side effects where the feature requires idempotency.
- Ownership and authorization must be checked server-side for protected data.

6. Acceptance Criteria

- Data model supports the required business relationships and constraints.
- API enforces authentication, authorization, validation and ownership where applicable.
- Business rules are server-authoritative.
- Transactions/idempotency/concurrency controls are used where the feature requires them.
- Controlled error codes and request correlation are implemented.

7. Test Cases

TC	Scenario	Expected Result
TC-S11-01	Expired	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

TC	Scenario	Expected Result
TC-S11-02	Future	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-03	Minimum not met	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-04	Maximum discount	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-05	User mismatch	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-06	Usage exhausted	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-07	Concurrent redemption	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-08	Duplicate apply	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S11-09	Over-discount attempt	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

8. Sprint Exit Checklist

- Implementation completed only for this platform.
- All primary and negative use cases pass.
- API contracts are implemented/consumed as defined.
- Validation and authorization tests pass.
- Loading/empty/error handling is verified where applicable.
- No critical or blocker defect remains.
- Automated tests pass in CI.
- Documentation and contract changes are committed.

9. Out of Scope for This Document

- Implementation behavior of Customer Portal. That behavior belongs in the separate S11 document for that platform.
- Implementation behavior of Admin Portal. That behavior belongs in the separate S11 document for that platform.